Primary Six Mathematics | Term 1 | Week 1 | Lesson 1: Types of Sets

What you will learn

• Recognise equal, unequal, equivalent and non-equivalent sets.

• Match elements of two sets correctly.

What is a set?

• A set is a well-defined collection of objects or numbers. The objects are called elements or members.

• Use curly brackets to list elements. For example, A = {2, 4, 6}. The number of elements is written n(A) = 3.

• Order does not matter in a set. Repeating an element does not increase the number of elements.

Equal and unequal sets

• Two sets are equal when they contain exactly the same elements. A = {1, 3, 5} and B = {5, 1, 3} are equal: A = B.

• Two sets are unequal if at least one element differs. C = {1, 3, 5} and D = {1, 3, 7} are unequal: C ≠ D.

Equivalent and non-equivalent sets

• Equivalent sets have the same number of distinct elements, even if the elements differ. E = {a, b, c} and F = {7, 8, 9} are equivalent because n(E) = n(F) = 3.

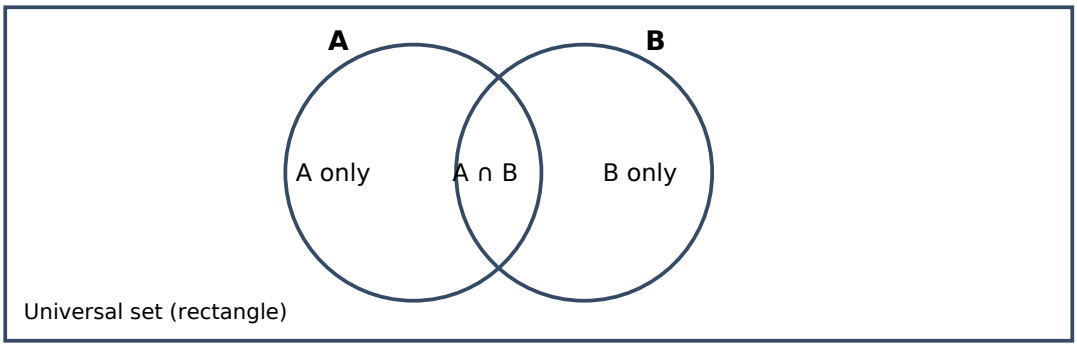
• Non-equivalent sets have different numbers of elements. G = {2, 4} and H = {p, q, r} are non-equivalent because 2 ≠ 3.

Matching sets

• Draw a line or an arrow from each element in one set to one element of another. This is one-to-one matching when no element is left over or matched twice.

• Example: {cup, plate, spoon} can match {1, 2, 3} in three one-to-one pairs.

Worked examples



1. Are {4, 6, 8} and {8, 6, 4} equal? Yes. Both contain 4, 6 and 8; their order is different but their elements are the same.

2. Are {m, n, o} and {10, 20, 30} equivalent? Yes. Each set has three elements.

3. If K = {1, 1, 2, 3}, what is n(K)? n(K) = 3. Repeated elements are counted once.

Practice

1. Write the elements of the set of even numbers greater than 0 and less than 10.

2. State whether {2, 5, 7} and {7, 2, 5} are equal. Give a reason.

3. State whether {a, e, i} and {1, 2, 3} are equivalent.

4. State whether {2, 4} and {2, 4, 6} are equivalent.

5. Write one set that is equal to {red, blue}.

6. If P = {3, 3, 5, 7, 7}, find n(P).

7. Give two unequal sets that are equivalent.

8. Are {1, 2, 3} and {1, 2, 4} equal? Are they equivalent?

Working space

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Primary Six Mathematics | Term 1 | Week 1 | Lesson 2: Intersecting, Disjoint, Union and Universal Sets

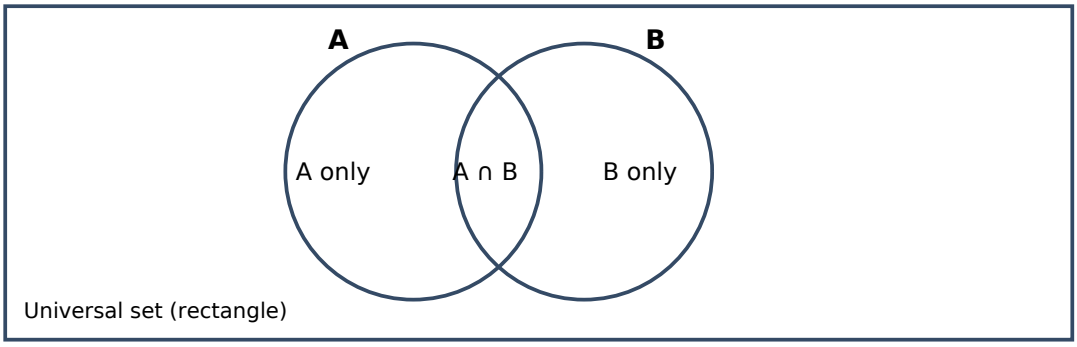
What you will learn

• Identify the intersection and union of two sets.

• Recognise disjoint and intersecting sets.

• Explain a universal set using a clear example.

Common elements: intersection



• The intersection A ∩ B contains elements found in both A and B.

• Example: A = {1, 2, 3, 4} and B = {3, 4, 5}. Then A ∩ B = {3, 4}.

• Sets with at least one common element are intersecting (or joint) sets.

Disjoint sets

• Disjoint sets have no common elements. Their intersection is the empty set, written ∅.

• Example: C = {2, 4, 6} and D = {1, 3, 5}. Then C ∩ D = ∅.

Union of sets

• The union A ∪ B contains all elements from A or B or both. List each element only once.

• For A = {1, 2, 3, 4} and B = {3, 4, 5}, A ∪ B = {1, 2, 3, 4, 5}.

Universal set

• The universal set, U, contains all elements being considered in a question.

• If we study whole numbers from 1 to 6, U = {1, 2, 3, 4, 5, 6}. A set in this question must use members from U.

Worked examples

1. Find {a, b, c} ∩ {b, d}. The only common member is b. Answer: {b}.

2. Find {1, 3} ∪ {3, 5, 7}. {1, 3, 5, 7}; write 3 only once.

3. Are {2, 6} and {1, 5} disjoint? Yes. They share no elements.

Practice

1. A = {2, 4, 6, 8}; B = {4, 8, 10}. Find A ∩ B.

2. Find A ∪ B for the sets in Question 1.

3. Are {a, b} and {c, d} disjoint?

4. Are {1, 2, 3} and {3, 5} intersecting?

5. Let U = {1, 2, 3, 4, 5}. Is {2, 5} a subset of U?

6. Find {red, green} ∩ {green, blue}.

7. Find {2, 4} ∪ ∅.

8. Draw two overlapping circles to show sets with a shared element 5.

Working space

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Primary Six Mathematics | Term 1 | Week 1 | Lesson 3: Difference and Complement of Sets

What you will learn

• Find the difference between two sets.

• Find the complement of a set relative to a stated universal set.

• Identify the correct Venn diagram regions.

Difference of sets

• $A - B$ means elements in A but not in B.

• If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$, then $A - B = \{1, 2\}$.

• $B - A = \{5\}$. Notice that $A - B$ is usually different from $B - A$.

Complement

• The complement of A, written A', contains elements in the universal set U that are not in A.

• If $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$, then $A' = \{1, 3, 5\}$.

• Always write or check the universal set before finding a complement.

Shading a Venn diagram

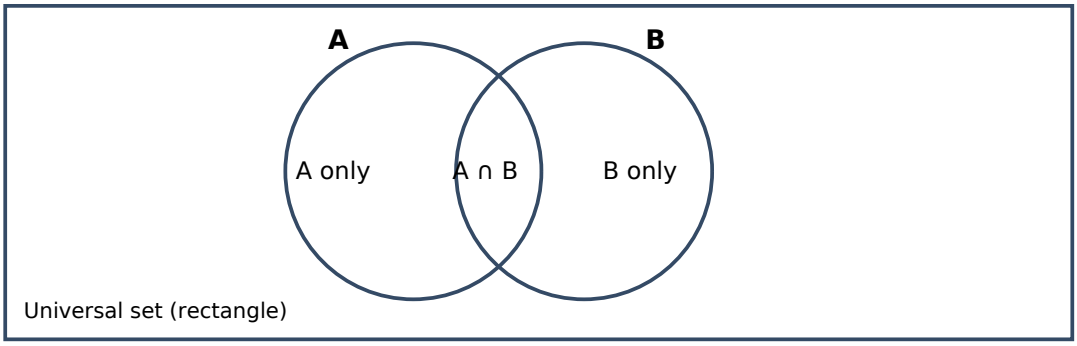
• To shade $A \cap B$, shade only the overlap.

• To shade $A \cup B$, shade both circles including the overlap.

• To shade $A - B$, shade the part of circle A outside circle B.

• To shade A', shade the region inside U but outside circle A.

Worked examples



1. $A = \{2, 3, 5, 7\}$; $B = \{3, 7, 9\}$. Find $A - B$. Remove 3 and 7 from A. Answer: $\{2, 5\}$.

2. With $U = \{a, b, c, d, e\}$ and $C = \{b, e\}$, find C'. $C' = \{a, c, d\}$.

3. Can 9 be in A' if 9 is not in U? No. A complement contains only elements from U.

Practice

1. $A = \{1, 2, 3, 4\}$; $B = \{3, 4, 5\}$. Find $A - B$.

2. For the same sets, find $B - A$.

3. Find A ∩ B for the sets above.

4. U = {1, 2, 3, 4, 5, 6}; C = {1, 3, 6}. Find C′.

5. U = {a, b, c, d}; D = {a, d}. Find D′.

6. Which region is shaded for A − B? Describe it in words.

7. If U = {2, 4, 6} and E = U, find E′.

8. Explain why a universal set is needed to find a complement.

Working space

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Primary Six Mathematics | Term 1 | Week 1 | Lesson 4: Subsets and the Number of Subsets

What you will learn

• Identify subsets and proper subsets.

• List the subsets of a small set.

• Find the number of subsets and proper subsets.

Subset

• A is a subset of B, written $A \subseteq B$, when every element of A belongs to B.

• Example: $\{2, 4\} \subseteq \{2, 4, 6\}$. Every set is a subset of itself.

• The empty set $\varnothing$ is a subset of every set.

Proper subset

• A proper subset contains some but not all elements of the original set. Write $A \subset B$ when A is a proper subset of B.

• Example: $\{2, 4\} \subset \{2, 4, 6\}$, but $\{2, 4, 6\}$ is not a proper subset of itself.

Listing subsets

• Let $A = \{a, b\}$. Its subsets are $\varnothing$, $\{a\}$, $\{b\}$, $\{a, b\}$.

• The proper subsets of A are $\varnothing$, $\{a\}$, $\{b\}$.

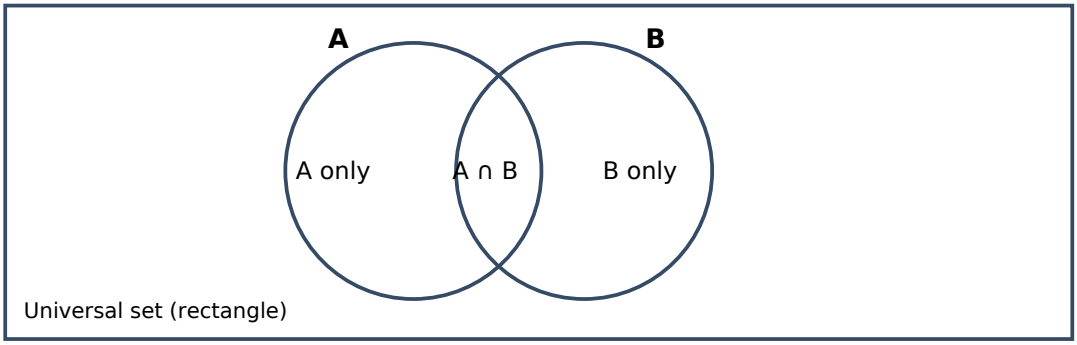
Counting subsets

• If a set has n distinct elements, the total number of subsets is 2^n.

• The number of proper subsets is $2^n - 1$.

• For a set of 3 elements: $2^3 = 8$ subsets and $8 - 1 = 7$ proper subsets.

Worked examples



1. List every subset of $\{1, 2\}$. $\varnothing$, $\{1\}$, $\{2\}$, $\{1, 2\}$.

2. How many subsets has a set of 4 elements? $2^4 = 16$ subsets.

3. How many proper subsets has a set of 5 elements? $2^5 - 1 = 32 - 1 = 31$.

Practice

1. Is {1, 3} a subset of {1, 2, 3, 4}?

2. List every subset of {p}.

3. List every subset of {x, y}.

4. Find the number of subsets of a set with 3 elements.

5. Find the number of proper subsets of a set with 3 elements.

6. Find the number of subsets of a set with 5 elements.

7. A set has 16 subsets. How many elements does it contain?

8. A set has 7 proper subsets. How many elements does it contain?

Working space

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Primary Six Mathematics | Term 1 | Week 1 | Lesson 5: Finding the Number of Elements in Sets

What you will learn

• Count distinct elements in a set.

• Use n(A) for the number of elements.

• Find the number of elements in a union, including overlap.

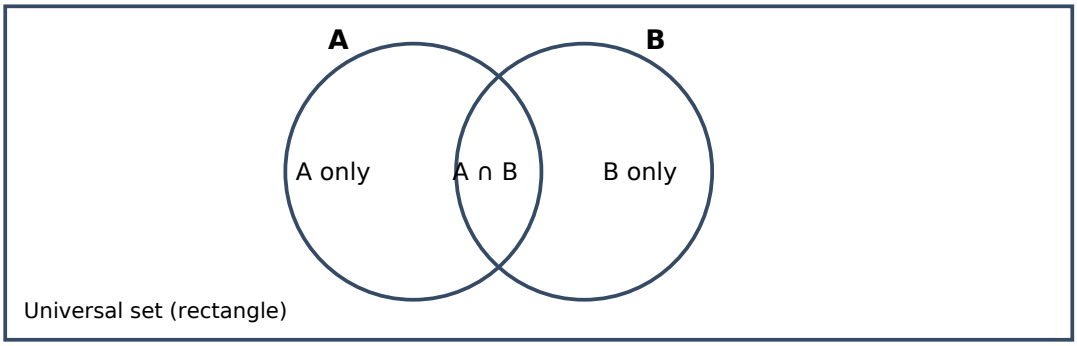
Counting elements

• The number of distinct elements in set A is written n(A).

• If A = {2, 4, 6, 8}, n(A) = 4. Repeated items are counted once: n({a, a, b}) = 2.

• An empty set has no elements: n(∅) = 0.

Elements in a Venn diagram



• A Venn diagram uses a circle for each set within a rectangle for the universal set.

• When circles overlap, the common members are placed only in the overlap. Do not count them twice.

Counting a union

• For two finite sets: n(A ∪ B) = n(A) + n(B) − n(A ∩ B).

• Example: n(A) = 9, n(B) = 7 and n(A ∩ B) = 3. Then n(A ∪ B) = 9 + 7 − 3 = 13.

• If A and B are disjoint, n(A ∩ B) = 0, so n(A ∪ B) = n(A) + n(B).

Counting regions

• Number in A only = n(A) − n(A ∩ B).

• Number in B only = n(B) − n(A ∩ B).

• Number outside both = total in U − n(A ∪ B), if U is given.

Worked examples

1. Find n({1, 2, 2, 3, 4, 4}). There are four distinct elements: 1, 2, 3, 4. So n(A) = 4.

2. n(A)=12, n(B)=10, n(A ∩ B)=4. Find n(A ∪ B). 12 + 10 − 4 = 18.

3. A class has 30 pupils; 18 like football, 13 like netball and 6 like both. How many like neither? At least one = 18 + 13 − 6 = 25. Neither = 30 − 25 = 5.

Practice

1. Find n({2, 3, 5, 7, 11}).

2. Find n({a, a, b, c, c}).

3. Find n(∅).

4. If n(A)=8, n(B)=6 and n(A ∩ B)=2, find n(A ∪ B).

5. If n(P)=15 and n(P ∩ Q)=4, find the number in P only.

6. A class has 40 pupils. 22 like mangoes, 19 like bananas and 8 like both. How many like at least one?

7. Using Question 6, how many like neither?

8. Two disjoint sets have 7 and 9 members. Find the number in their union.

Working space

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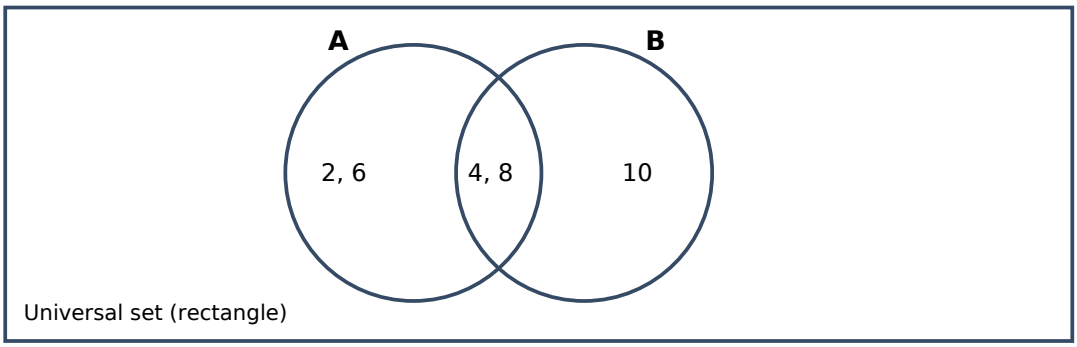
Primary Six Mathematics — Representing and interpreting Venn diagrams

Name: _____ Date: _____

Learn and understand

A Venn diagram uses circles to show groups. Each item belongs in the correct region. Put common items in the overlapping part first.

Worked examples



Example 1: A={2,4,6,8}, B={4,8,10}. Fill the diagram.

Working: Both: {4,8}; A only: {2,6}; B only: {10}.

Example 2: Find n(A∪B).

Working: Count each member once: {2,4,6,8,10}, so n(A∪B)=5.

Practice independently

Show your working and units.

1. A={1,2,3}, B={2,3,4}. List the overlap.

2. List A only.

3. List B only.

4. Find n(A∪B).

5. A={a,b,c}, B={c,d}. List the union.

6. Find n(A∩B).

7. Draw two circles for sets of odd and prime numbers from 1 to 10.

8. Why is 2 in the prime-only region?

Check your understanding

Explain the rule you used: _____

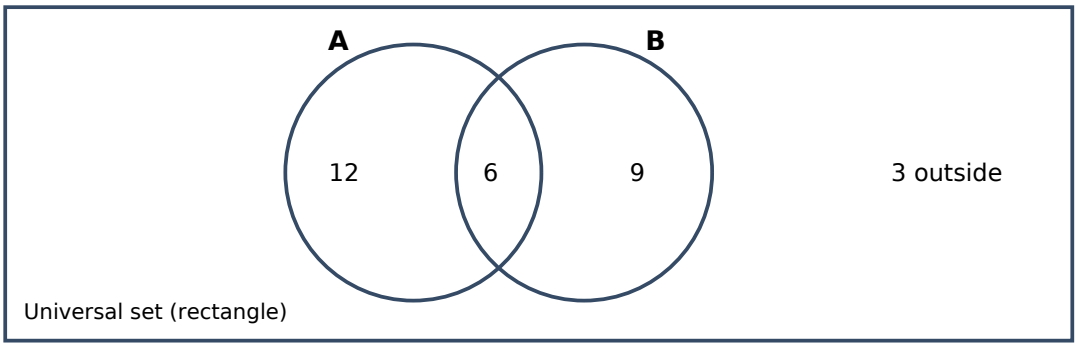
Primary Six Mathematics — Word problems with two sets

Name: _____ Date: _____

Learn and understand

For two sets, count the shared group once. n(A∪B)=n(A)+n(B)−n(A∩B). If everybody is included, the union is the class total.

Worked examples



Example 1: In 30 pupils, 18 play netball, 15 play football and 6 play both. How many play at least one?

Working: 18+15−6=27. Neither=30−27=3.

Example 2: If 24 like tea, 20 like milk and 10 like both, how many like only tea?

Working: 24−10=14.

Practice independently

Show your working and units.

1. In a class, 14 like rice, 12 like matooke and 5 like both. Find the union.

2. How many like rice only?

3. How many like matooke only?

4. In a group of 30, 21 study Art, 18 study Music, 12 study both. Find neither.

5. If n(A)=13, n(B)=16 and n(A∩B)=4, find n(A∪B).

6. If union=28, n(A)=20, n(B)=17, find intersection.

7. Write the number in exactly one group if A only=7 and B only=11.

8. Why must you subtract the intersection when adding group totals?

Check your understanding

Explain the rule you used: _____

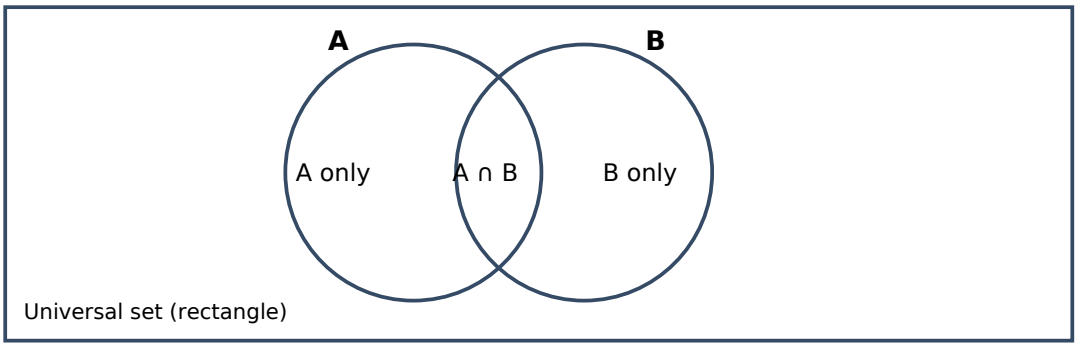
Primary Six Mathematics — Simple probability

Name: _____ Date: _____

Learn and understand

Probability tells how likely an event is. For equally likely outcomes, probability = favourable outcomes ÷ total outcomes. An impossible event has probability 0 and a certain one has probability 1.

Worked examples



Example 1: A fair die is thrown. Find P(even).

Working: Even faces: 2,4,6. P=3/6=1/2.

Example 2: A bag contains 3 red and 5 blue beads. Find P(red).

Working: 3 out of 8, so P=3/8.

Practice independently

Show your working and units.

1. P(head) for a fair coin.

2. P(5) on a fair six-sided die.

3. P(number greater than 4) on the die.

4. A bag has 4 green and 6 yellow beads. P(green)?

5. Find P(yellow).

6. P(7) on an ordinary die?

7. P(a number from 1 to 6) on an ordinary die?

8. If P(rain)=2/5, find P(no rain).

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Place value and digit value

Name: _____ Date: _____

Learn and understand

Place value names the position of a digit. Digit value means its worth in that position. Read big numbers in groups of three digits.

Worked examples

Example 1: In 4,725,306 find the place and value of 2.

Working: The 2 is in the ten-thousands place; its value is 20,000.

Example 2: Find the value of 7 in 7,092,145.

Working: 7,000,000.

Practice independently

Show your working and units.

1. Value of 6 in 963,102.

2. Place of 8 in 4,817,293.

3. Value of 3 in 503,214.

4. What digit is in the millions place in 8,657,300?

5. Value of 5 in 1,530,041.

6. Write 40,000+2,000+300+8 as a numeral.

7. Which is greater, 506,070 or 560,007?

8. Find the difference between digit value and place value.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Expanded whole numbers and powers of ten

Name: _____ Date: _____

Learn and understand

Expanded form shows the sum of the values of each digit. Powers of ten use exponents: $10^2=100$ and $10^3=1{,}000$.

Worked examples

Example 1: Expand 52,304.

Working: $50{,}000+2{,}000+300+4$.

Example 2: Write 3,406 in powers of ten.

Working: $3\times10^3+4\times10^2+0\times10^1+6\times10^0$.

Practice independently

Show your working and units.

1. Expand 6,205.

2. Expand 91,047.

3. Find 10^4.

4. Find 4×10^3.

5. Write 7,030 as a sum.

6. Write $8\times10^2+5\times10+2$.

7. What is 10^0?

8. Find $2\times10^4+3\times10^2+7$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Standard form or scientific notation

Name: _____ Date: _____

Learn and understand

A large or small number can be written $a \times 10^n$, where $1 \le a < 10$ and n is an integer. Moving a decimal point left gives a positive exponent for large numbers.

Worked examples

Example 1: Write 42,000 in standard form.

Working: 4.2×10^4.

Example 2: Write 0.0035 in standard form.

Working: 3.5×10^{-3}.

Practice independently

Show your working and units.

1. 8,000 in standard form.

2. 63,000 in standard form.

3. 0.004 in standard form.

4. 0.00072 in standard form.

5. 3.6×10^3 as a numeral.

6. 5×10^{-2} as decimal.

7. 1,250,000 in standard form.

8. Explain the range of a in standard form.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Writing expanded numbers as numerals

Name: _____ Date: _____

Learn and understand

To find a numeral from expanded form, multiply each digit by its place value and add the amounts. Keep zero places.

Worked examples

Example 1: Find $5\times10^4+2\times10^3+6\times10+8$.

Working: $50{,}000+2{,}000+60+8=52{,}068$.

Example 2: Find $300{,}000+40{,}000+50+9$.

Working: $340{,}059$.

Practice independently

Show your working and units.

1. $6{,}000+400+30+2$.

2. $9\times10^3+3\times10^2+5$.

3. $70{,}000+800+6$.

4. $2\times10^5+4\times10^3+3\times10$.

5. $8\times10^2+2\times10+1$.

6. $5\times10^4+7$.

7. $900{,}000+9{,}000+90+9$.

8. $6\times10^3+0\times10^2+4\times10+2$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Reading and writing whole numbers in words

Name: _____ Date: _____

Learn and understand

Separate a large number into millions, thousands and units. Use British English spelling and hyphens for compound tens.

Worked examples

Example 1: Write 2,304,015 in words.

Working: Two million, three hundred and four thousand and fifteen.

Example 2: Write 70,021 in words.

Working: Seventy thousand and twenty-one.

Practice independently

Show your working and units.

1. Write 8,001 in words.

2. Write 20,020 in words.

3. Write 304,005 in words.

4. Write 1,000,000 in words.

5. Write 42,019 in words.

6. Write 900,300 in words.

7. Write 14,070 in words.

8. Write 5,050,500 in words.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Changing number words to figures

Name: _____ Date: _____

Learn and understand

Read the million and thousand groups first. Put three digits in each group; use zeros for missing hundreds or tens.

Worked examples

Example 1: Write “three million, forty thousand and six” in figures.

Working: 3,040,006.

Example 2: Write “six hundred and two thousand and seventy” in figures.

Working: 602,070.

Practice independently

Show your working and units.

1. Eight thousand and nine.

2. Seventy thousand and fifty.

3. Three hundred and ten thousand.

4. Two million, six thousand and four.

5. Nine hundred thousand and five.

6. Twenty-five thousand, one hundred and four.

7. Four million and thirty.

8. Five hundred and six thousand, seven hundred.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Rounding whole numbers

Name: _____ Date: _____

Learn and understand

Identify the requested place. Look at the digit immediately to its right: 0–4 means keep the digit; 5–9 means increase it by one. Replace following digits with zero.

Worked examples

Example 1: Round 47,682 to the nearest thousand.

Working: Hundreds digit is 6, so 48,000.

Example 2: Round 635,124 to nearest ten thousand.

Working: Thousands digit is 5, so 640,000.

Practice independently

Show your working and units.

1. 368 to nearest ten.

2. 4,249 to nearest hundred.

3. 7,950 to nearest thousand.

4. 82,499 to nearest ten thousand.

5. 75 to nearest ten.

6. 14,501 to nearest thousand.

7. 999 to nearest hundred.

8. 6,544 to nearest ten.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Decimal place values

Name: _____ Date: _____

Learn and understand

Digits after a decimal point show tenths, hundredths and thousandths. In 4.507, the 5 means five tenths, the 0 means zero hundredths and the 7 means seven thousandths.

Worked examples

Example 1: Find the value of 8 in 6.084.

Working: 8 hundredths = 0.08.

Example 2: Write 3+2/10+5/100 as decimal.

Working: 3.25.

Practice independently

Show your working and units.

1. Value of 4 in 12.405.

2. Value of 7 in 0.078.

3. Place of 9 in 5.009.

4. 3+6/10+2/100.

5. Write 0.5 as a fraction.

6. Write 0.04 as a fraction.

7. Find 4 ones+6 hundredths.

8. Compare 0.7 and 0.65.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Writing decimals in words and figures

Name: _____ Date: _____

Learn and understand

Read a decimal as whole units and a fraction of a unit. A final digit in the hundredths position means “hundredths”.

Worked examples

Example 1: Write 6.25 in words.

Working: Six and twenty-five hundredths.

Example 2: Write “three and seven thousandths” as a decimal.

Working: 3.007.

Practice independently

Show your working and units.

1. Write 0.6 in words.

2. Write 2.04 in words.

3. Nine and five tenths in figures.

4. Four and eight hundredths in figures.

5. Seven thousandths in figures.

6. Write 10.01 in words.

7. One and twelve hundredths in figures.

8. Write 0.205 in words.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Expanded form of decimals

Name: _____ Date: _____

Learn and understand

Split a decimal into ones, tenths, hundredths and thousandths. The decimal-place values are 1/10, 1/100 and 1/1000.

Worked examples

Example 1: Expand 7.438.

Working: 7+0.4+0.03+0.008.

Example 2: Expand 52.06.

Working: 50+2+0.06.

Practice independently

Show your working and units.

1. Expand 3.25.

2. Expand 0.407.

3. Expand 15.08.

4. Expand 2.006.

5. Express 4.32 using fractions.

6. Expand 0.91.

7. Expand 8.005.

8. Expand 70.304.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Finding decimals from expanded form

Name: _____ Date: _____

Learn and understand

Add the parts by aligning decimal places. A zero keeps a place empty; do not omit it from the middle.

Worked examples

Example 1: Find 5+0.2+0.007.

Working: 5.207.

Example 2: Find 30+4+0.06+0.003.

Working: 34.063.

Practice independently

Show your working and units.

1. 2+0.4+0.05.

2. 8+0.003.

3. 0.6+0.08.

4. 10+0.9+0.004.

5. 7+5/100.

6. 1+4/10+6/1000.

7. 20+0.01+0.008.

8. 3+0.7+0.02+0.005.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Ordering decimals

Name: _____ Date: _____

Learn and understand

Align decimal places and compare digits from left to right. Adding zeros to the right does not change a decimal.

Worked examples

Example 1: Order 0.9, 0.85 and 0.805 from smallest.

Working: 0.805, 0.85, 0.9 (write 0.900 for comparison).

Example 2: Which is greater, 3.07 or 3.7?

Working: 3.7 because 3.70>3.07.

Practice independently

Show your working and units.

1. Ascending: 0.2, 0.12, 0.21.

2. Descending: 4.04,4.4,4.004.

3. Compare 0.80 and 0.8.

4. Compare 1.09 and 1.1.

5. Smallest: 0.06,0.6,0.606.

6. Largest: 2.003,2.03,2.3.

7. Ascending: 5.1,5.01,5.001.

8. Insert < or >: 0.507 __ 0.57.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Rounding decimals

Name: _____ Date: _____

Learn and understand

Choose the decimal place. Inspect the following digit. If it is 5 or more, round up; otherwise leave the digit unchanged.

Worked examples

Example 1: Round 5.678 to 2 decimal places.

Working: Third decimal is 8: 5.68.

Example 2: Round 9.46 to 1 decimal place.

Working: Hundredths digit is 6: 9.5.

Practice independently

Show your working and units.

1. 3.74 to 1 dp.

2. 3.75 to 1 dp.

3. 0.986 to 2 dp.

4. 4.005 to 2 dp.

5. 8.499 to whole number.

6. 8.5 to whole number.

7. 12.347 to 2 dp.

8. 0.444 to 2 dp.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Roman numerals

Name: _____ Date: _____

Learn and understand

Common symbols are I=1, V=5, X=10, L=50, C=100, D=500 and M=1,000. A smaller symbol before a larger one can mean subtraction, as in IV=4.

Worked examples

Example 1: Write 49 in Roman numerals.

Working: XLIX = 40+9.

Example 2: Convert CDXLVI to figures.

Working: 400+40+6=446.

Practice independently

Show your working and units.

1. 19 in Roman numerals.

2. 40 in Roman numerals.

3. 90 in Roman numerals.

4. 99 in Roman numerals.

5. 144 in Roman numerals.

6. MM in figures.

7. DCC in figures.

8. CMIV in figures.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Operations with Roman numerals

Name: _____ Date: _____

Learn and understand

Convert Roman numerals to ordinary figures, carry out the operation, then change the answer back to Roman numerals.

Worked examples

Example 1: Calculate XII+IX.

Working: 12+9=21=XXI.

Example 2: Calculate XL−XIV.

Working: 40−14=26=XXVI.

Practice independently

Show your working and units.

1. V+VI.

2. XX−VIII.

3. X+XV.

4. L−X.

5. IX+IX.

6. C−L.

7. XL+X.

8. XXX−XIX.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Counting in base two and base five

Name: _____ Date: _____

Learn and understand

A base shows how many digits are used. Base two uses 0,1; base five uses 0,1,2,3,4. Each column is a power of the base.

Worked examples

Example 1: Change 1011_2 to base ten.

Working: $1\times8+0\times4+1\times2+1=11$.

Example 2: Change 23_5 to base ten.

Working: $2\times5+3=13$.

Practice independently

Show your working and units.

1. 10_2 in decimal.

2. 111_2 in decimal.

3. 100_2 in decimal.

4. 14_5 in decimal.

5. 32_5 in decimal.

6. 20_5 in decimal.

7. Write 5 in base two.

8. Write 8 in base five.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Adding numbers in other bases

Name: _____ Date: _____

Learn and understand

Add from the right. In base five, a total of five makes a carry of one to the next column; in base two a total of two makes a carry.

Worked examples

Example 1: Add 23_5+14_5.

Working: $3+4=7=12_5$: put 2, carry 1; $2+1+1=4$; answer 42_5.

Example 2: Add 101_2+11_2.

Working: $5+3=8=1000_2$.

Practice independently

Show your working and units.

1. 12_5+11_5.

2. 24_5+1_5.

3. 101_2+1_2.

4. 11_2+11_2.

5. 13_5+12_5.

6. 110_2+10_2.

7. 21_5+3_5.

8. $100_2 + 101_2$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Subtracting and multiplying in other bases

Name: _____ Date: _____

Learn and understand

Borrow a group of the base, not ten. Check each answer by changing both numbers into base ten.

Worked examples

Example 1: Subtract $43_5 - 14_5$.

Working: $23 - 9 = 14$ decimal, which is 24_5.

Example 2: Multiply $12_5 \times 3_5$.

Working: $7 \times 3 = 21$ decimal $= 41_5$.

Practice independently

Show your working and units.

1. $24_5 - 3_5$.

2. $30_5 - 1_5$.

3. $110_2 - 10_2$.

4. $101_2 - 11_2$.

5. $4_5 \times 2_5$.

6. $11_2 \times 10_2$.

7. $12_5 \times 2_5$.

8. $100_2 \times 11_2$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Finite number systems and clock arithmetic

Name: _____ Date: _____

Learn and understand

In a finite cycle numbers repeat after a fixed number of steps. On a 12-hour clock, 11 hours after 4 o’clock is 3 o’clock.

Worked examples

Example 1: What is 9+7 on a 12-hour clock?

Working: 16; subtract 12, giving 4.

Example 2: A 7-day cycle begins Monday. What day is 10 days later?

Working: 10 leaves remainder 3 after 7; Thursday.

Practice independently

Show your working and units.

1. On 12-hour clock, 8+6.

2. On 12-hour clock, 11+3.

3. On 12-hour clock, 2+10.

4. On 7-day cycle, two days after Friday.

5. On 7-day cycle, seven days after Tuesday.

6. On 5-step cycle labelled 0–4, 3+4.

7. On 5-step cycle labelled 0–4, 4+4.

8. Why do remainders help?

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Subtracting with finite cycles

Name: _____ Date: _____

Learn and understand

Move backward around the cycle to subtract. On a 12-hour clock, subtracting 5 hours from 3 o’clock gives 10 o’clock.

Worked examples

Example 1: On a 12-hour clock, calculate 2−7.

Working: Go back 7 hours from 2: 7 o’clock.

Example 2: In a 7-day cycle, what day was four days before Wednesday?

Working: Saturday.

Practice independently

Show your working and units.

1. 12-hour clock: 4−6.

2. 12-hour clock: 9−11.

3. 12-hour clock: 5−3.

4. 12-hour clock: 1−2.

5. 3 days before Monday.

6. 5 days before Thursday.

7. On 5-step cycle 0–4: 1−3.

8. On 5-step cycle 0–4: 2−4.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Unknowns in finite systems

Name: _____ Date: _____

Learn and understand

When a clock equation has an unknown, test the position that gives the required result after full cycles. Equivalent clock times differ by the cycle size.

Worked examples

Example 1: Find x between 1 and 12 if x+5 gives 3 on a 12-hour clock.

Working: x+5=15, so x=10.

Example 2: Find x in {0,1,2,3,4} if x+4 gives 2 in a 5-step cycle.

Working: x=3 because 3+4=7, remainder 2.

Practice independently

Show your working and units.

1. 12-hour cycle: x+3=2.

2. 12-hour cycle: x+7=4.

3. 12-hour cycle: x−5=2.

4. 5-step cycle: x+2=0.

5. 5-step cycle: x+1=4.

6. 7-step cycle 0–6: x+3=1.

7. 12-hour cycle: x+9=12.

8. 5-step cycle: $x-2=4$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Applications of finite cycles

Name: _____ Date: _____

Learn and understand

Use cycles for days of the week, months, clock faces and repeating patterns. Divide the number of steps by the length of the cycle and use the remainder.

Worked examples

Example 1: Today is Thursday. What day will it be in 17 days?

Working: 17=14+3: Sunday.

Example 2: A clock shows 9 o’clock. What time will it show 28 hours later?

Working: 28=24+4: 1 o’clock.

Practice independently

Show your working and units.

1. 10 days after Monday.

2. 15 days after Friday.

3. 24 hours after 6 o’clock.

4. 14 days before Sunday.

5. 49 days after Wednesday.

6. 20 hours after 8 o’clock.

7. 9 days before Tuesday.

8. 30 hours after 5 o’clock.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Adding whole numbers

Name: _____ Date: _____

Learn and understand

Align digits by place value. Start at the units column and carry a ten to the next column when needed.

Worked examples

Example 1: Add 38,745+26,589.

Working: 38,745+26,589=65,334.

Example 2: A trader sells 2,475 oranges on Monday and 1,988 on Tuesday. How many?

Working: 2,475+1,988=4,463 oranges.

Practice independently

Show your working and units.

1. 468+257.

2. 2,345+1,876.

3. 87,600+9,405.

4. 625+375.

5. 39,900+101.

6. 14,325+2,500.

7. 12,345+54,321.

8. 3,067+2,908.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Subtracting whole numbers

Name: _____ Date: _____

Learn and understand

Write digits under matching place values. Exchange from the column to the left when the upper digit is smaller.

Worked examples

Example 1: Subtract 28,476−9,858.

Working: 18,618.

Example 2: A school has 1,250 exercise books and gives out 785. How many remain?

Working: 1,250−785=465 books.

Practice independently

Show your working and units.

1. 700−258.

2. 5,000−1,999.

3. 43,105−2,678.

4. 1,204−589.

5. 90,000−800.

6. 726−447.

7. 35,100−12,905.

8. $6{,}302 - 3{,}408$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Multiplication of whole numbers

Name: _____ Date: _____

Learn and understand

Break a multiplier into hundreds, tens and ones. Multiply by each place and add the partial products.

Worked examples

Example 1: Multiply 325×56.

Working: 325×(50+6)=16,250+1,950=18,200.

Example 2: A box holds 24 pencils. How many in 36 boxes?

Working: 24×36=864 pencils.

Practice independently

Show your working and units.

1. 45×23.

2. 126×14.

3. 208×35.

4. 125×8.

5. 304×12.

6. 67×19.

7. 250×24.

8. 111×111.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Long division of whole numbers

Name: _____ Date: _____

Learn and understand

Divide from left to right. Multiply back, subtract and bring down the next digit. Confirm by multiplying quotient by divisor.

Worked examples

Example 1: Divide 3,684 by 12.

Working: 12×307=3,684, so quotient 307.

Example 2: Share 1,248 mangoes equally among 24 shops.

Working: 1,248÷24=52 mangoes per shop.

Practice independently

Show your working and units.

1. 936÷12.

2. 1,008÷24.

3. 1,440÷36.

4. 805÷5.

5. 2,304÷48.

6. 4,200÷70.

7. 2,475÷25.

8. 6,048÷12.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Division word problems

Name: _____ Date: _____

Learn and understand

Find what is being shared or grouped, the number in each group and the unit of the answer. Divide and check by multiplication.

Worked examples

Example 1: A school shares 1,176 books equally among 28 classes. Find books per class.

Working: 1,176÷28=42 books.

Example 2: A farmer packs 975 eggs in trays of 25. How many trays?

Working: 975÷25=39 trays.

Practice independently

Show your working and units.

1. 540 learners travel equally in 12 buses. Learners per bus?

2. 756 pencils shared among 21 pupils. Each gets?

3. 840 kg divided into 35 sacks. kg per sack?

4. 1,000 shillings shared equally by 8 children. Each gets?

5. 720 books packed 24 per carton. Number of cartons?

6. 1,260 litres into 42 drums. Each drum?

7. How many groups of 15 in 435?

8. A farmer has 924 seedlings in 22 rows. Seedlings per row?

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Mixed operations and BODMAS

Name: _____ Date: _____

Learn and understand

Follow brackets, orders (powers), division/multiplication from left to right, then addition/subtraction from left to right.

Worked examples

Example 1: Calculate $8+6\times3$.

Working: Multiply first: $8+18=26$.

Example 2: Calculate $(18-6)\div3+2^2$.

Working: $12\div3+4=8$.

Practice independently

Show your working and units.

1. $4+5\times6$.

2. $(4+5)\times6$.

3. $24\div6\times2$.

4. $36-8\div2$.

5. 3^2+7.

6. $20-(6+4)$.

7. $5\times(3+2)-4$.

8. $12+18\div3-2$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Number operation properties

Name: _____ Date: _____

Learn and understand

Commutative: change order in addition/multiplication. Associative: change grouping. Distributive: multiply each part of a sum.

Worked examples

Example 1: Use distributive property for $7\times(20+3)$.

Working: $7\times20+7\times3=140+21=161$.

Example 2: Use associative property for $(4\times25)\times3$.

Working: $4\times(25\times3)=300$.

Practice independently

Show your working and units.

1. $8+5=5+___$.

2. $6\times9=9\times___$.

3. $3\times(10+4)$.

4. $(2+3)+7=2+(3+___)$.

5. $5\times(8+2)$.

6. $4\times(20-3)$.

7. Can subtraction be commutative?

8. Name property: $2\times(3+4)=2\times3+2\times4$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Divisibility tests

Name: _____ Date: _____

Learn and understand

A number is divisible by 2 if its last digit is even; by 5 if it ends in 0 or 5; by 10 if it ends in 0; by 3 or 9 if its digit sum is divisible by 3 or 9.

Worked examples

Example 1: Is 4,356 divisible by 9?

Working: 4+3+5+6=18, which is divisible by 9, so yes.

Example 2: Is 2,148 divisible by 4?

Working: Last two digits 48 divisible by 4, so yes.

Practice independently

Show your working and units.

1. Is 345 divisible by 3?

2. Is 345 divisible by 9?

3. Is 2,160 divisible by 10?

4. Is 2,160 divisible by 8?

5. Is 234 divisible by 6?

6. Is 1,025 divisible by 5?

7. Is 148 divisible by 4?

8. Is 237 divisible by 2?

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Developing number patterns

Name: _____ Date: _____

Learn and understand

Look for a fixed difference, a multiplication rule or a shape pattern. Triangular numbers count objects arranged in triangles; square numbers count square arrays.

Worked examples

Example 1: Continue 3,6,10,15, __.

Working: Differences 3,4,5, so next difference 6: 21.

Example 2: Continue 2,4,8,16, __.

Working: Multiply by 2: 32.

Practice independently

Show your working and units.

1. 2,5,8,11,___.

2. 1,4,9,16,___.

3. 1,3,6,10,___.

4. 40,35,30,___.

5. 3,9,27,___.

6. 4,8,12,16,___.

7. 1,2,4,7,11,___.

8. 100,50,25,___.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Prime and composite numbers

Name: _____ Date: _____

Learn and understand

A prime number has exactly two positive factors: 1 and itself. A composite number has more than two. The number 1 is neither prime nor composite.

Worked examples

Example 1: Is 29 prime?

Working: Yes. It has factors 1 and 29 only.

Example 2: Is 21 composite?

Working: Yes. $21=3\times7$, so it has more than two factors.

Practice independently

Show your working and units.

1. Is 2 prime?

2. Is 1 prime?

3. Is 9 composite?

4. List primes below 10.

5. Is 31 prime?

6. Is 49 composite?

7. Smallest composite number?

8. Is 37 prime?

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Consecutive whole numbers

Name: _____ Date: _____

Learn and understand

Consecutive whole numbers come one after another: n, n+1, n+2. Use equations when the sum is known.

Worked examples

Example 1: Three consecutive whole numbers sum to 48. Find them.

Working: Let them be n,n+1,n+2. 3n+3=48; n=15. Numbers 15,16,17.

Example 2: Find two consecutive numbers whose sum is 29.

Working: n+(n+1)=29; n=14, so 14 and 15.

Practice independently

Show your working and units.

1. Numbers after 19 and 20.

2. Three consecutive numbers beginning at 7.

3. Sum of 8,9,10.

4. Two consecutive numbers sum to 51.

5. Three consecutive numbers sum to 36.

6. Write three consecutive numbers after n.

7. Find n if n+(n+1)=17.

8. Are 5,7,9 consecutive whole numbers?

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Consecutive odd and even numbers

Name: _____ Date: _____

Learn and understand

Consecutive odd numbers differ by 2, and consecutive even numbers also differ by 2. Use n,n+2,n+4.

Worked examples

Example 1: Three consecutive even numbers sum to 42.

Working: n+(n+2)+(n+4)=42; n=12. Numbers 12,14,16.

Example 2: Two consecutive odd numbers sum to 40.

Working: n+(n+2)=40; n=19, giving 19,21.

Practice independently

Show your working and units.

1. Next two evens after 18.

2. Next two odds after 21.

3. Sum of 5,7,9.

4. Two consecutive evens sum to 26.

5. Two consecutive odds sum to 32.

6. Three consecutive even numbers sum to 60.

7. Three consecutive odd numbers sum to 45.

8. Difference between consecutive odd numbers.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Factors and highest common factor

Name: _____ Date: _____

Learn and understand

A factor divides a whole number exactly. Common factors divide both numbers. The highest common factor (HCF/GCF) is the largest common factor.

Worked examples

Example 1: Find HCF of 24 and 36.

Working: Factors of 24: 1,2,3,4,6,8,12,24. Factors of 36 include 1,2,3,4,6,9,12,18,36. HCF=12.

Example 2: Find HCF of 18 and 30.

Working: Common factors 1,2,3,6; HCF=6.

Practice independently

Show your working and units.

1. Factors of 10.

2. HCF of 12 and 18.

3. HCF of 15 and 25.

4. HCF of 7 and 9.

5. HCF of 16 and 24.

6. HCF of 21 and 28.

7. Is 6 a factor of 42?

8. Largest factor of 29?

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Prime factorisation

Name: _____ Date: _____

Learn and understand

Break a composite number into prime factors. You may use a factor tree or repeated division. Write repeated factors with exponents.

Worked examples

Example 1: Prime-factorise 72.

Working: $72=8\times9=2\times2\times2\times3\times3=2^3\times3^2$.

Example 2: Prime-factorise 90.

Working: $90=2\times3\times3\times5=2\times3^2\times5$.

Practice independently

Show your working and units.

1. Prime factors of 12.

2. Prime factors of 20.

3. Prime factors of 36.

4. Prime factors of 50.

5. Prime factors of 48.

6. Prime factors of 100.

7. Prime factors of 27.

8. Prime factors of 84.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Finding numbers from prime factors

Name: _____ Date: _____

Learn and understand

To recover a number from its prime factorisation, evaluate each power and multiply.

Worked examples

Example 1: Find $2^3 \times 3^2$.

Working: $8 \times 9 = 72$.

Example 2: Find $2^2 \times 5^2$.

Working: $4 \times 25 = 100$.

Practice independently

Show your working and units.

1. 2×3^2.

2. $2^3 \times 5$.

3. $3^2 \times 7$.

4. $2^4 \times 3$.

5. $5^2 \times 2$.

6. $2^2 \times 3 \times 5$.

7. $3^3 \times 2$.

8. 2×7^2.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Multiples and lowest common multiple

Name: _____ Date: _____

Learn and understand

Multiples of a number come from repeated multiplication. The lowest common multiple (LCM) is the smallest positive multiple shared by two or more numbers.

Worked examples

Example 1: Find LCM of 6 and 8.

Working: Multiples of 6: 6,12,18,24; of 8: 8,16,24. LCM=24.

Example 2: Find LCM of 9 and 12.

Working: 9:9,18,27,36; 12:12,24,36. LCM=36.

Practice independently

Show your working and units.

1. LCM of 4 and 6.

2. LCM of 5 and 10.

3. LCM of 6 and 9.

4. LCM of 3 and 7.

5. LCM of 8 and 12.

6. LCM of 7 and 14.

7. LCM of 10 and 15.

8. List first three multiples of 9.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — HCF and LCM using prime factors

Name: _____ Date: _____

Learn and understand

For HCF, use the lowest power of each shared prime. For LCM, use the highest power of every prime appearing in either number.

Worked examples

Example 1: Find HCF and LCM of 12 and 18.

Working: $12=2^2\times3$; $18=2\times3^2$. HCF$=2\times3=6$; LCM$=2^2\times3^2=36$.

Example 2: Find HCF and LCM of 20 and 30.

Working: $20=2^2\times5$; $30=2\times3\times5$. HCF$=10$; LCM$=60$.

Practice independently

Show your working and units.

1. HCF,LCM of 8 and 12.

2. HCF,LCM of 6 and 15.

3. HCF,LCM of 9 and 12.

4. HCF,LCM of 10 and 25.

5. HCF,LCM of 14 and 21.

6. HCF,LCM of 16 and 20.

7. HCF,LCM of 18 and 24.

8. Check HCF×LCM for 12 and 18.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Finding unknown prime factors

Name: _____ Date: _____

Learn and understand

Use factor trees, shared prime factors and the identity HCF(a,b)×LCM(a,b)=a×b for positive whole numbers.

Worked examples

Example 1: Two numbers are 12 and x. Their HCF is 6 and LCM 60. Find x.

Working: 6×60=12×x; x=30.

Example 2: If $36=2^2\times3^2$, which prime factor is missing in 2×2×3×__?

Working: 3.

Practice independently

Show your working and units.

1. Find x if HCF=3, LCM=24, a=6.

2. Find x if HCF=5, LCM=60, a=15.

3. Missing factor: 30=2×3×__.

4. Missing factor: 84=2×2×3×__.

5. HCF of 18 and 27.

6. LCM of 18 and 27.

7. HCF×LCM for 8 and 12.

8. Find b if HCF=4, LCM=40, a=8.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Applying HCF and LCM

Name: _____ Date: _____

Learn and understand

Use HCF for the greatest equal grouping without leftovers. Use LCM for the first time repeating events coincide.

Worked examples

Example 1: Ropes of 24 m and 36 m must be cut into longest equal pieces. Length?

Working: HCF(24,36)=12 m.

Example 2: Two bells ring every 6 and 8 minutes. When together next?

Working: LCM(6,8)=24 minutes.

Practice independently

Show your working and units.

1. Longest equal pieces from 18 m and 30 m.

2. Bells every 4 and 10 min: next together.

3. HCF of 16 and 40.

4. LCM of 9 and 15.

5. Maximum equal groups from 24 and 32 items.

6. Bus visits every 12 and 18 days: next same day.

7. HCF of 28 and 42.

8. LCM of 5 and 6.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Word problems with common multiples

Name: _____ Date: _____

Learn and understand

Use a timeline for repeating events. Find LCM of the intervals; units must match before calculating.

Worked examples

Example 1: A bus comes every 15 min and a van every 20 min. Both come at 8:00. Next together?

Working: LCM(15,20)=60 min, so 9:00.

Example 2: Lights flash every 8 s and 12 s. When together again?

Working: LCM(8,12)=24 s.

Practice independently

Show your working and units.

1. Bells every 6 and 9 min.

2. Buses every 10 and 25 min.

3. Lights every 4 and 7 s.

4. Two cycles 12 and 15 days.

5. Bells every 5 and 8 min.

6. Two clocks every 20 and 30 min.

7. Lights every 6 and 10 seconds.

8. Intervals 9 and 12 min starting 10:00. Next same time?

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Powers of whole numbers

Name: _____ Date: _____

Learn and understand

An exponent tells how many times the base is used as a factor. For the same base, multiply powers by adding exponents and divide powers by subtracting exponents (nonzero base).

Worked examples

Example 1: Simplify $2^3 \times 2^4$.

Working: $2^7 = 128$.

Example 2: Simplify $5^4 \div 5^2$.

Working: $5^2 = 25$.

Practice independently

Show your working and units.

1. 3^2.

2. 2^5.

3. 4^3.

4. 10^3.

5. $2^2 \times 2^3$.

6. $3^4 \div 3^2$.

7. 5^0.

8. $(2^2)^3$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Squares of whole numbers and fractions

Name: _____ Date: _____

Learn and understand

To square a number, multiply it by itself. For fractions, square numerator and denominator. A decimal square can be found by multiplication.

Worked examples

Example 1: Find $(3/4)^2$.

Working: 9/16.

Example 2: Find 1.2^2.

Working: $1.2 \times 1.2 = 1.44$.

Practice independently

Show your working and units.

1. 6^2.

2. 11^2.

3. $(2/3)^2$.

4. $(1/5)^2$.

5. 0.5^2.

6. 1.5^2.

7. 15^2.

8. $(1\ 1/2)^2$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Square roots of whole numbers

Name: _____ Date: _____

Learn and understand

A square root is a number whose square gives the original number. A perfect square has a whole-number square root.

Worked examples

Example 1: Find √144.

Working: 12×12=144, so √144=12.

Example 2: Find √225.

Working: 15×15=225, so √225=15.

Practice independently

Show your working and units.

1. √25.

2. √64.

3. √81.

4. √121.

5. √169.

6. √196.

7. √400.

8. √1.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Square roots of fractions and decimals

Name: _____ Date: _____

Learn and understand

Write an exact square fraction as a square root of its numerator over its denominator. For decimals, use place value or change to a fraction.

Worked examples

Example 1: Find √(9/16).

Working: √9/√16=3/4.

Example 2: Find √0.49.

Working: 0.7×0.7=0.49; answer 0.7.

Practice independently

Show your working and units.

1. √(1/4).

2. √(25/36).

3. √0.09.

4. √0.64.

5. √(49/100).

6. √2.25.

7. √(4/9).

8. √0.01.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Applying square roots

Name: _____ Date: _____

Learn and understand

Square roots help find the side of a square when its area is known. Keep units consistent: square metres for area, metres for length.

Worked examples

Example 1: A square field has area 196 m^2. Find its side.

Working: $\sqrt{196}=14$ m.

Example 2: A square tile has area 0.81 m^2. Find the side.

Working: $\sqrt{0.81}=0.9$ m.

Practice independently

Show your working and units.

1. Square area 64 cm^2: side.

2. Square area 225 m^2: side.

3. Square area 121 cm^2: side.

4. Square area 0.25 m^2: side.

5. Square side 7 cm: area.

6. Square perimeter 40 m: side.

7. Square area 400 cm^2: perimeter.

8. Side 12 m: area.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Cubes and cube roots

Name: _____ Date: _____

Learn and understand

A cube is a number multiplied by itself three times. The cube root undoes cubing. For example, $4^3=64$ and $\sqrt[3]{64}=4$.

Worked examples

Example 1: Find 5^3.

Working: $5\times5\times5=125$.

Example 2: Find $\sqrt[3]{216}$.

Working: $6^3=216$, so $\sqrt[3]{216}=6$.

Practice independently

Show your working and units.

1. 2^3.

2. 3^3.

3. 4^3.

4. 10^3.

5. $\sqrt[3]{8}$.

6. $\sqrt[3]{27}$.

7. $\sqrt[3]{125}$.

8. $\sqrt[3]{1000}$.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Sequences and number rules

Name: _____ Date: _____

Learn and understand

A sequence is an ordered list that follows a rule. Identify the rule and use it to find the missing or next terms.

Worked examples

Example 1: Find the next two terms: 5,10,20,40,...

Working: Multiply by 2: 80,160.

Example 2: Fill 2,6,12,20,___.

Working: Differences 4,6,8, so next difference 10: 30.

Practice independently

Show your working and units.

1. 7,14,21,___.

2. 81,27,9,___.

3. 2,5,10,17,___.

4. 1,8,27,___.

5. 12,17,22,___.

6. 3,6,12,24,___.

7. 100,90,80,___.

8. 1,4,7,10,___.

Check your understanding

Explain the rule you used: _____

Primary Six Mathematics — Number puzzles and magic squares

Name: _____ Date: _____

Learn and understand

A 3×3 magic square uses each number 1–9 once. Every row, column and diagonal has sum 15. Use arithmetic and checking to fill blanks.

Worked examples

Example 1: Find missing entry in a row: 8,1,__ where sum is 15.

Working: 15−8−1=6.

Example 2: Why is 5 in the centre of the usual 1–9 magic square?

Working: The middle entry balances the rows and diagonals; the standard arrangement has 5 at the centre.

Practice independently

Show your working and units.

1. Complete a row: 4,9,__ to total 15.

2. Complete a row: 3,5,__ to total 15.

3. Complete a row: 2,7,__ to total 15.

4. Sum of digits 1 through 9.

5. Magic sum for the normal 3×3 square.

6. Centre of the normal 3×3 square.

7. Check 8+1+6.

8. What must be true of every main diagonal?

Check your understanding

Explain the rule you used: _____